

ABHISHEK M R

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Portfolio: [abhishekmr.vercel.app](#) | GitHub: [github.com/abhi-byte62](#) | LinkedIn: [in/abhishekmr029](#) | LeetCode: [u/playboldAbhi](#)

PROFESSIONAL SUMMARY

Software Engineer with a strong Computer Science foundation specializing in backend architectures, distributed systems, and low-latency C++/Java applications. Experienced in developing deterministic limit order book matching engines, event-driven microservices with RabbitMQ, stream backpressure pipelines, and optimistic concurrency control systems. Proficient in Data Structures, Algorithms, System Design, and building verified, high-throughput software.

TECHNICAL SKILLS

Languages:	C++17, Java 21, JavaScript (ES6+), TypeScript, Python 3, C, SQL
Backend & Distributed:	Spring Boot 3, Node.js, Express, FastAPI, RabbitMQ, WebSockets, RESTful APIs
Databases & Caching:	PostgreSQL 16/17, MySQL, Redis 7 (Pub/Sub & Caching), Prisma ORM
Systems & Architecture:	Limit Order Books (LOB), Fixed-Point Math, Stream Backpressure, Concurrency, Zero-Copy Buffers, WebGL
Infrastructure & Tools:	Docker & Compose, Git, GitHub Actions, Linux / POSIX Shell, Postman, Vitest
Core CS Foundations:	Data Structures & Algorithms, OOP, OS (Memory, I/O), DBMS, Computer Networks, System Design, OCC

SELECTED ENGINEERING PROJECTS

StackLens — Website Architecture Inference & Intelligence Engine | Java 21, Spring Boot 3, RabbitMQ, PostgreSQL, Redis, React 19, TypeScript

[\[GitHub\]](#) [\[Demo\]](#)

- Architected an asynchronous multi-vector probe engine reverse-engineering public web applications across 200+ weighted signatures.
- Implemented SSRF and DNS-rebinding perimeter defense by validating resolved target IPs against private RFC 1918 subnets prior to worker dispatch.
- Decoupled long-running external HTTP/TLS crawlers from client APIs using RabbitMQ message queues and Redis caching tiers.
- Synthesized PostgreSQL DDL schemas and REST/GraphQL API contracts from detected access patterns with interactive DAG topology generation.

LiquidityLens — Market Microstructure Simulator & Matching Engine | C++17, Python 3.10, FastAPI, Fixed-Point Math, Hawkes Processes, WebSockets

[\[GitHub\]](#) [\[Demo\]](#)

- Developed a deterministic C++17 limit order book (LOB) matching engine supporting 4.5M+ events/s with ~220ns match latency.
- Utilized int64_t fixed-point arithmetic across matching and accounting pipelines, eliminating IEEE-754 floating-point drift.
- Simulated clustered order arrival dynamics with synthetic Hawkes processes and modeled FIFO queue priority across 10μs–500μs latency slips.
- Calculated zero look-ahead adverse selection markout curves at 10ms, 100ms, and 1s post-trade horizons; streamed L2 depth via WebSockets.

TaskFlow — Real-Time Collaborative Kanban & Distributed State Engine | React 18, Node.js, PostgreSQL 17, Prisma ORM, Socket.io, TanStack Query

[\[GitHub\]](#) [\[Demo\]](#)

- Engineered a real-time collaborative workspace with Optimistic Concurrency Control (OCC) using integer revision tags to reject stale writes.
- Implemented fractional midpoint indexing for O(1) card drag reordering, preventing expensive cascading updates across database rows.
- Integrated room-scoped Socket.io state synchronization with role-based access control (RBAC) enforced in atomic Prisma transactions.

Specter Proxy — Stream Backpressure & Ephemeral TLS Interception Proxy | Node.js Streams, HTTP/HTTPS, TLS Termination, Dynamic SNI, Backpressure

[\[GitHub\]](#) [\[Demo\]](#)

- Built a stream-oriented forward proxy for real-time packet inspection, latency injection, and synthetic network resilience testing.
- Enforced strict highWaterMark stream backpressure flow control to prevent heap bloat, maintaining a bounded ~35MB memory footprint.
- Implemented dynamic on-the-fly X.509 SNI certificate generation signed by an ephemeral local CA for seamless TLS MITM inspection.

EDUCATION

Presidency University — Bengaluru, India

Sept 2023 – Present

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Expected 2027

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems (DBMS), Computer Networks, System Design, Software Engineering.

PROBLEM SOLVING & CERTIFICATIONS

- **Competitive Programming:** Active problem solver on [LeetCode \(u/playboldAbhi\)](#) and [Codeforces \(playboldAbhi\)](#), focusing on Graph Algorithms, Dynamic Programming, and Amortized Complexity.
- **HackerRank Certifications:** SQL (Advanced), Rest API (Intermediate), Problem Solving (Intermediate), Java (Basic).